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Neurological diseases at presymptomatic stages

Artificial intelligence in presymptomatic neurological diseases: Bridging normal variation and prodromal signatures



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ABSTRACT

Presymptomatic neurological diseases are marked by early pathological changes that occur before overt clinical symptoms. These stages, which include prodromes such as REM sleep behavior disorder in Parkinson's or mild cognitive impairment in Alzheimer's, offer critical opportunities for early intervention. However, their detection remains challenging due to the subtlety of changes and the overlap with normal interindividual variability. Artificial intelligence (AI), especially machine learning (ML) and deep learning (DL), offers new tools to uncover hidden signatures in complex biomedical data. First, we explore how supervised ML models can detect known prodromal patterns across diverse modalities, including EEG, cognitive scores, and structural imaging. Depending on the input, various model types — such as tree-based algorithms for structured data and convolutional or transformer networks for images and signals — can extract predictive features of early neurodegeneration. These approaches have demonstrated success in identifying at-risk individuals before clinical thresholds are reached. Yet, detecting only known patterns limits the scope of early intervention. Many individuals who will go on to develop neurological disease may not yet exhibit any recognized prodromal syndrome. Bridging this gap requires moving beyond predefined labels toward models capable of identifying subtle, unknown anomalies in individuals still considered healthy. Second, we address the detection of latent anomalies among individuals not yet considered at risk without identifiable known prodromal patterns. By mining clinical records, free-text medical notes, and population-level health

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databases (e.g., UK Biobank, EDS-AP-HP), and by analyzing sensor data from smartphones or wearables, AI can flag deviations from healthy patterns long before symptom onset or formal diagnosis. This approach holds promise for scalable, low-burden, ecological screening. Finally, we introduce the concept of pseudo-healthy twins — synthetic, personalized baselines generated from structural data such as MRI, to improve anomaly detection. These models predict a patient's expected healthy signal in another modality, such as PET, enabling the subtraction of normal anatomical and physiological variability to isolate disease-specific effects. Generative models like GANs and VAEs have shown promise in producing these cross-modal references, enhancing early anomaly detection in diseases like Alzheimer's and multiple sclerosis. Together, these approaches show how AI can bridge the gap between normal variation and early pathology, enabling more sensitive, personalized, and population-scalable detection of presymptomatic neurological disease.

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1. Introduction

Presymptomatic neurological diseases are characterized by pathological changes in the nervous system that occur before full diagnostic criteria are met. This stage is considered presymptomatic either because the patient does not yet experience any noticeable symptoms, or because clinical examination and diagnostic tools are not sensitive enough to detect objective signs of the disease. While the presymptomatic stage refers to the presence of disease without overt symptoms, like the radiologically isolated syndrome in multiple sclerosis [1], the prodromal stage is characterized by early, subtle signs that precede full diagnosis. Such prodromal patterns have been described in several neurological diseases — for example, rapid eye movement (REM) sleep behavior disorder as a prodrome of Parkinson's disease [2], and mild cognitive impairment in Alzheimer's disease [3]. Identifying such presymptomatic and prodromal cases is crucial for enabling early intervention, improving long-term outcomes, and including patients in therapeutic trials that may alter the course of the disease [1,2]. However, by definition, the frequent lack of diagnostic criteria or clear definition of the pre-symptomatic stages makes their detection particularly challenging for the clinician, a challenge that could potentially be addressed with the help of artificial intelligence (AI), which excels at uncovering subtle patterns in complex biomedical data beyond human perception.

AI refers to the development of computer systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, decision-making, perception, and language understanding, and is increasingly transforming the biomedical field [4–6]. At the core of AI lie machine learning and its subfield, deep learning [6]. Machine learning encompasses a range of techniques that enable algorithms to learn from data and improve their performance through experience, by automatically identifying patterns and making decisions rather than relying on fixed, rule-based instructions for each task. These are typically categorized as either supervised or unsupervised learning. In supervised learning, the model is trained on labeled data, where the expected outputs are known — for example, using

clinical data to predict disease presence. Common supervised algorithms include support vector machines, decision trees, random forests, and deep neural networks. In unsupervised learning, the algorithm explores unlabeled data to uncover hidden patterns or structures — for instance, grouping patients based on similar imaging or genomic features. Techniques like k-means clustering, principal component analysis, and autoencoders can be used in this context [6].

Deep learning, a specialized subset of machine learning, relies on multi-layered artificial neural networks to model complex and high-dimensional data [6]. These networks are inspired by the structure of the human brain and are composed of interconnected units called artificial neurons, which process and transmit information through weighted connections known as artificial synapses. One particularly important class of deep learning models in biomedicine is the convolutional neural network (CNN), especially effective for image analysis tasks. By applying learnable filters across input images, CNNs are able to automatically detect and extract hierarchical features — such as edges, textures, or shapes — making them highly effective for visual pattern recognition and image-based biomedical tasks such as lesion detection or organ segmentation in medical imaging [6]. AI applications in healthcare have also extended to natural language processing, particularly in the context of the emergence of large language models (4), which involves analyzing unstructured clinical texts such as radiology reports or electronic health records to extract relevant medical information and support clinical workflows.

The tasks that AI can address in the biomedical field are diverse. Arguably, three major applications include: classification (e.g., distinguishing patients from healthy controls using imaging data [5]), image segmentation (e.g., delineating anatomical structures or regions of interest in scans [7]), and image synthesis (e.g., generating realistic images in a different imaging modality [8]). Thanks to their ability to learn rich hierarchical representations, deep learning models are particularly well suited for detection tasks where the patterns of interest may be subtle or diffuse, such as in the early identification of disease in presymptomatic individuals.

First, we will explore how AI can enhance sensitivity and automate the detection of prodromal patterns — allowing us to identify individuals who may already be on the path to illness, even if they do not yet meet established diagnostic criteria. Second, we will examine how AI, when combined with ecological sensors, can uncover abnormalities of unknown significance in individuals generally considered healthy, thereby identifying those potentially at risk. Finally, we will discuss how AI can bridge the gap between normal inter-individual variability and the early detection of the very earliest subtle signs of disease by generating “pseudo-healthy twins” — synthetic models that serve as personalized baselines for comparison and more precise anomaly detection.

2. Automated detection of known prodromal patterns using AI: insights across modalities and models

Features of early, presymptomatic neurological patterns — such as radiologically isolated syndrome —, and prodromal neurological patterns — such as REM sleep behavior disorder, or mild cognitive impairment — can be detected using machine learning models trained on large annotated datasets. These datasets typically compare individuals who remain healthy (Healthy Controls, HCs) with those who later develop a defined neurological disease [5].

Different types of input data (or modalities) require different types of AI models. When the input is tabular data — like clinical, demographic, or neuropsychological scores — classic machine learning models are well suited and, in particular, powerful extensions of tree-based models (e.g. random forests, gradient boosting) are still often then state of the art [9].

For time-series data like electroencephalography (EEG) or electromyography (EMG), where signals vary over time and are often noisy, deep learning becomes particularly useful. In these cases, the goal is to identify subtle, sometimes invisible, patterns that may suggest early neural dysfunction. This can be done either by using signal pre-processing techniques (like time-frequency analysis to extract suspicious events such as interictal grapho-elements in EEG), or directly using models that can learn from raw signals. For example, CNNs can automatically extract low-level features (like spikes or rhythms) in early layers, and more complex, abstract patterns in deeper layers [10]. These networks have also been applied to full polysomnography recordings, combining EEG, EMG, electrooculography (EOG), and more, to detect early markers of neurodegeneration [11]. They can also perform multimodal fusion of these inputs to extract specific patterns between them.

In medical imaging, CNNs are still the standard for many tasks. They are especially powerful in identifying features within magnetic resonance imaging (MRI) or computed tomography scans by learning spatial patterns. For instance, in multiple sclerosis, an enlarged choroid plexus has been identified as a potential biomarker even in patients with radiologically isolated syndrome [12]. CNNs enable automated segmentation of affected structures [7], making these techni-

ques accessible and scalable to clinical cohorts [13]. Deep learning is particularly effective for tasks such as segmentation and image synthesis, where each image provides a large quantity of supervising information (each pixel/voxel is label). However, the situation is different for tasks such as classification and regression in which each image contributes only one label. In such case, for typical samples sizes found in medicine (hundreds to a few thousands), linear models are often as good as deep learning ones [14]. This situation is very different from that of computer vision where one typically has millions on training imaging.

Transformer architectures [15] have led to huge improvements in performance across many fields, from natural language processing to computer vision. This has also had an important impact on medical AI [4]. Unlike CNNs, which process images in a hierarchical and localized way (assuming nearby pixels are more related), Transformers analyze the entire input simultaneously. This global perspective allows them to better capture long-range dependencies and relationships within the data, making them particularly powerful for detection tasks where distant regions of an image interact [4].

3. Detection of latent anomalies in ecological settings among clinically normal individuals

While the previously described AI models are designed to recognize known prodromal patterns from various input signals, a more ambitious goal is to extend this capability to individuals who do not present with any of these early warning syndromes — those typically considered healthy. In this context, AI could help detect early, subtle abnormalities during routine primary care visits. The feasibility of these approaches lies in the availability for model training of large-scale national care databases which combine medical records, diagnostic information, medicines, and paraclinical examinations [6], such as the Health Data Hub (<https://www.health-data-hub.fr/>), the *Entrepôt des Données de Santé* like the one from the Assistance Publique des Hôpitaux de Paris (<https://eds.aphp.fr/>), or the UK Biobank [16].

Interestingly, some non-specific symptoms — such as fatigue, mild depressive states, or constipation — frequently lead to primary care consultations. While these symptoms are common and often benign, population-level studies have shown that such consultations, especially when linked to patterns of medication use and healthcare reimbursement data, can occur more frequently years before the diagnosis of neurodegenerative diseases like multiple sclerosis or Alzheimer's disease [17,18].

By applying deep learning approaches to natural language processing — particularly large language models (LLMs) — in the context of medical records, it may be possible to detect distinctive features in clinical notes from early, seemingly unrelated visits. One of the most valuable sources of information in this context is the free-text medical notes written by the clinician. Just as LLMs can be trained to recognize harmful or inappropriate content from subtle cues in language [19], they could also be trained to detect subtle indicators within clinical narratives that suggest a higher risk

of an underlying presymptomatic condition. In practice, this could mean that an AI system flags a consultation for fatigue or low mood as potentially prodromal, prompting the healthcare provider to recommend additional evaluations — such as an MRI or further specialist referral — even in the absence of overt neurological signs.

While individuals presenting to primary care with vague complaints such as fatigue or low mood may not meet criteria for a defined prodromal syndrome, they are, by definition, not entirely asymptomatic, as evidenced by their decision to seek primary care. However, AI's potential goes even further: it may one day allow for the identification of at-risk individuals who are completely asymptomatic — those who have never sought medical care and therefore have no formal clinical history or paraclinical examinations on record. For these individuals, input data must come from ecological sources — that is, data collected passively or actively in daily life — as, by definition, no medical data is available.

The widespread availability of digital sensors in smartphones and wearables offers a largely untapped resource for capturing ecological data, and recent studies have shown that, with the help of AI, these devices can extract digital biomarkers that reflect subtle changes in neurological health [6]. Although few of these studies have specifically focused on presymptomatic cases so far, the approach holds strong potential for early detection and monitoring. Several types of digital sensors have already been successfully explored in this context. For example, smartphones have been used by Cohen and colleagues [20], who developed the MS Screen Test app to detect subtle neurological differences between healthy controls and individuals with radiologically isolated syndrome, capturing biomarkers related to motor speed, coordination, vision, and cognition through short, sensitive tasks. Voice recordings, also collected via smartphones, have been used by Rahmatallah and colleagues [21], who showed that CNNs analyzing voice spectrograms could accurately detect Parkinson's disease. This approach highlights the potential of smartphones as non-invasive, scalable tools for remote neurological screening. Smartwatches represent another promising sensor. Ko and colleagues [22] demonstrated that accelerometry and heart rate data collected during sleep, when analyzed through machine learning algorithms, can detect abnormal REM sleep behavior and altered sleep stages in Parkinson's disease patients. In terms of movement analysis, wearable joint accelerometers were employed by Borzi and colleagues [23], who developed a model to detect freezing of gait episodes in Parkinson's disease, enabling robust, home-based gait assessment. Similarly, camera-based systems have been used by Kushioka and colleagues [24], who applied computer vision and deep learning to analyze video-recorded gait for the diagnosis of locomotive syndrome, using only a single fixed camera. Extending beyond structured clinical tasks, the concept of the digital phenotype [25] introduces a broader view: how individuals' passive interactions with technology can reveal health-relevant behavioral patterns. For instance, long-term analysis of insomnia-related tweet patterns has shown potential for identifying sleep disturbances and mental health conditions, using social media as a passive sensor [25]. Although still emerging in psychiatry [26], digital phenotype analysis also holds immense

potential in neurology, where it could enable unobtrusive detection of subtle changes in presymptomatic stages.

These ubiquitous, low-burden tools make it possible to monitor individuals continuously in their natural environments, far beyond the spatial and temporal constraints of clinical settings. This is particularly valuable for presymptomatic individuals — those living with an undetected condition who may not seek medical attention until the disease is well advanced. For this population, only data from everyday life — captured through microphones, accelerometers, or sleep trackers — can reveal subtle deviations from typical patterns. When integrated with powerful AI models, these accessible digital sensors offer a promising path to democratizing early detection: enabling presymptomatic identification that is personalized, and scalable at the population level. By comparing sensor-derived ecological data — such as that collected from wearables like smartwatches — to large normative datasets continuously gathered from the general population using the same types of devices, AI can detect subtle deviations from healthy patterns, even in individuals who appear entirely asymptomatic. Much like how smartwatches now detect arrhythmias and prompt cardiology consultations, similar tools in neurology could serve as early-warning systems, guiding users toward timely evaluation and intervention.

Moreover, combining data from multiple modalities — multimodal data fusion — can significantly enhance the robustness of AI predictions. For example, an ambiguous MRI finding (such as nonspecific white matter lesions) may not offer enough diagnostic certainty on its own. However, when fused with ecological data — such as abnormal gait patterns or disrupted sleep — AI can construct a more confident, holistic risk profile.

From a methodological perspective, while the primary goal of these AI systems may be general anomaly detection — identifying deviations from normative behavior across a population, similar to how smartphones flag abnormal ECG patterns — this approach also holds the potential to uncover novel, disease-specific prodromal patterns. By analyzing large-scale ecological data from individuals who appear healthy but later develop neurological conditions, AI models could retrospectively learn which subtle behavioral, physiological, or sensor-derived changes consistently precede specific pathologies. Over time, what begins as broad anomaly detection could evolve into data-driven discovery of new prodromal signatures, enriching our understanding of early disease trajectories and improving the specificity of presymptomatic screening. One notable example of this approach is the UK Biobank accelerometry study [27]. Between 2013 and 2016, around 100,000 participants wore wrist monitors for a week, generating rich movement and sleep data. Ongoing follow-ups and extensive phenotyping make it possible to link these sensor metrics to imaging, mental health, and neurodegenerative conditions like Alzheimer's.

In summary, anomalies detected through everyday sensor data could lead individuals to seek medical care, where non-specific primary care consultations may be further informed by these ecological signals to guide additional investigations, such as MRI.

4. AI-driven pseudo-healthy twin generation: modeling healthy variability to improve early anomaly detection

One of the main challenges in detecting early signs of disease in non-symptomatic, clinically normal individuals lies in the natural variability among healthy subjects. This is particularly relevant in the context of neurodegenerative diseases like Alzheimer's disease, which typically affects older adults. In this population, significant inter-individual differences in cortical atrophy — both local and global —, due to normal aging and other diseases, can occur independently of any Alzheimer's disease -specific pathological process [28], making it difficult to isolate specific disease-related features.

For example, one of the earliest detectable alterations in AD is a reduction in functional synapse density, which can be measured using fluorodeoxyglucose positron emission tomography computed tomography (FDG-PET) imaging. However, FDG-PET signals are also sensitive to structural changes such as cortical atrophy [29]. This complicates interpretation, as hypometabolism observed on PET may reflect structural variation rather than true functional impairment [30].

This relationship can be conceptually expressed as:

$$\text{Observed signal} = \text{disease effect} + \text{anatomical and physiological variability}$$

To improve specificity in anomaly detection, the goal is to isolate the disease effect from the observed signal. To do this, one strategy is to model and subtract the anatomical component, specific to each patient:

$$\text{Disease effect} = \text{observed signal} - \text{anatomical and physiological variability}$$

The key challenge, then, is how to estimate this anatomical and physiological variability. One approach involves using an anatomical image — such as a T1-weighted MRI (T1w) — and learning from existing datasets how this anatomy typically maps onto another modality, like FDG-PET, in healthy individuals. This allows the generation of a synthetic FDG-PET image based on the patient's anatomy — effectively creating a personalized, pseudo-healthy twin for comparison (Fig. 1).

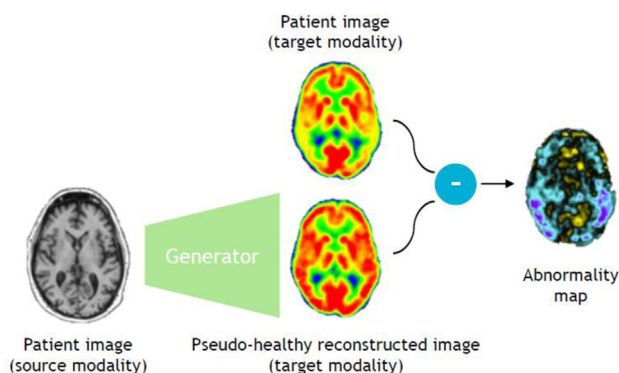


Fig. 1 – Pseudo-healthy twin of the subject reconstruction by deep learning to highlight abnormality with respect to the real modality taking into account anatomy-explained signal variations.

Early methods relied on atlas-based approaches, which showed that regions identified through such subtractions correlated with cognitive deficits [31]. However, atlas-based models can struggle to capture the complex and individual-specific relationships between structure and function.

With advances in deep learning, generative models — particularly those based on CNNs — offer a more powerful way to capture these relationships. For instance, in epilepsy, Flaus and colleagues [8] used generative adversarial networks (GANs) to synthesize normal FDG-PET images from T1-weighted MRI inputs. By subtracting the synthetic PET from the actual PET scan, they successfully highlighted local hypometabolism, offering a precise map of potential epileptogenic foci. These GANs must first be trained on datasets where both MRI and PET images are available for each subject, allowing the network to learn cross-modal correspondences.

Going further, Hassanal and colleagues [32] demonstrated the use of variational autoencoders (VAEs) to generate a subject's expected healthy signal without relying directly on anatomical inputs. This is especially interesting, as it shows that VAEs can model the gap between healthy controls and patients by learning how disease distorts normal physiological patterns.

These generative approaches, while they require further improvement and validation, could ultimately be integrated into the clinical routine of nuclear medicine physicians to enhance the interpretation of PET-MRI in patients with mild cognitive impairment, helping to detect early, subtle signs of Alzheimer's disease by comparing actual data to a personalized, AI-generated healthy baseline.

Moreover, amyloid PET tracers such as [11C]-PiB have been shown to bind not only to amyloid plaques but also to myelin [33]. Building on this, recent AI methods have been developed to translate structural MRI into synthetic amyloid PET images, allowing estimation of myelin-related signal variations in multiple sclerosis [34]. A similar approach could be applied in Alzheimer's disease, using these generative algorithms to produce a synthetic "pseudo-healthy twin" — that is, a PET image predicted from the individual's MRI, capturing only the non-amyloid (primarily myelin-related) component of the PET signal. By subtracting this synthetic reference from the actual amyloid PET scan, it may be possible to isolate the specific signal attributable to amyloid plaques.

These two approaches — based on FDG PET and amyloid PET — could even be combined in the same patient, enabling the simultaneous detection of anomalies in both metabolism and amyloid deposition, and providing a more comprehensive view of the early disease process.

5. Conclusion – perspectives

By leveraging large-scale health data, AI offers a powerful framework for the early and personalized identification of presymptomatic individuals through the detection of prodromal syndromes, latent anomalies in real-world conditions, and the creation of pseudo-healthy twins to isolate disease-specific signals. Integrating multimodal data — such as neuroimaging, genetic markers, cognitive profiles, and digital biomarkers — AI can enable risk stratification, guide early

intervention, and support the identification of prognostic factors or treatment conditions associated with slower disease progression. Beyond individual prediction, this approach facilitates the recruitment of well-characterized cohorts for therapeutic trials and may offer new insights into the natural history of neurological diseases, particularly in their earliest and most subtle stages.

However, integrating these tools into clinical practice remains challenging. Beyond technical hurdles — such as model validation, standardization, and integration into care workflows — the use of AI in asymptomatic individuals raises profound ethical challenges. Predicting future neurological decline in otherwise healthy people disrupts conventional care models and poses risks of anxiety, stigmatization, and discrimination, especially in the absence of effective treatments. This uncertainty may erode patient trust and make individuals reluctant to engage with AI-driven risk assessments. To be clinically and socially acceptable, early detection must be accompanied by clear communication, protective regulation, and concrete benefits for patients.

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